

INSUREAI

Evidence-Grounded Insurance Pitch Generator and Audit System

1. Project Overview

INSUREAI is an AI-based insurance advisory prototype designed to help advisors generate client-specific insurance pitch presentations using information from policy documents.

The system takes a company and its insurance priorities as input, compares selected insurance policies, generates a recommendation, and verifies important claims against the original policy brochures before they are presented to the client.

The main objective is to make AI-generated insurance recommendations more grounded, traceable, and reviewable rather than relying only on generated text.

2. Approach

The application follows the workflow:

Client Input → Policy Retrieval → AI Recommendation → Claim Verification → Advisor Review → Client Pitch & Audit Report

First, the advisor enters the company and its insurance priorities and selects the policies to compare.

Relevant information is retrieved from the available policy brochures and supplied as context to the AI model. The model then generates a recommendation based on the client requirements and available policy evidence.

Important claims generated in the recommendation are separately checked against the retrieved policy evidence. Each claim is classified according to its verification result.

The advisor can then review the generated recommendation and audit information before approving the final client-facing output.

3. Evidence-Grounded Generation

A key feature of INSUREAI is that recommendations are connected to the supplied insurance documents.

Instead of allowing the AI model to freely generate insurance benefits, the application retrieves relevant sections from policy brochures and uses them as supporting context. The generated claims are then checked against source evidence.

The audit report records information such as:

- Generated claim
- Verification status
- Source document
- Source page
- Citation match

- Retrieval similarity
- Supporting evidence
- Audit reason

For the sample TATA pitch, 4 claims were audited and all 4 were classified as supported, resulting in a 100% verification rate.

For example, the system verified Niva Bupa's air-ambulance coverage of up to INR 2,50,000 per hospitalisation against page 2 of the corresponding brochure.

4. Advisor Review and Guardrails

INSUREAI is designed as an advisor-support system, not an automatic policy-selection system.

Before the final presentation is exported, the advisor can review the recommendation and its supporting evidence.

The application also avoids presenting unsupported information as confirmed fact. Where sufficient evidence is unavailable, the exported presentation can indicate that the information requires advisor review rather than inventing a value.

The final recommendation remains advisory and requires review of policy terms, exclusions, pricing, and insurer quotations before client use. This is also explicitly stated in the generated presentation.

5. Tools and Technologies

The prototype uses a web-based frontend and Python backend.

- Frontend: React / TypeScript with TanStack-based application components.
- Backend: Python and FastAPI for API endpoints, workflow execution, document processing, recommendation generation, auditing, and exports.
- AI/LLM: Groq-hosted language model for generating evidence-grounded insurance recommendations and responses.
- Document Retrieval / RAG: Policy brochure content is retrieved and supplied to the language model so that generated insurance information is based on the available documents.
- PowerPoint Generation: python-pptx is used to populate a predefined presentation template and create the client pitch.
- PDF Generation: The application generates a separate claim audit report for review and submission.

6. Sample Output

The sample demonstration was created for TATA.

The generated presentation recommends Niva Bupa for consideration and compares it with HDFC ERGO. The recommendation uses documented benefits including air-ambulance coverage and in-patient care, including AYUSH.

The final presentation contains five slides:

1. Company overview and workforce considerations

2. Recommended policy and supporting benefits
3. Other policy considered
4. Policy comparison
5. Final recommendation rationale

The accompanying audit report provides claim-level traceability back to the original insurance policy documents.

7. Key Design Decisions

The system was designed around three main principles: grounding, traceability, and human review.

- Grounding reduces the possibility of unsupported insurance claims by using policy documents as the information source.
- Traceability allows an advisor to understand where an AI-generated claim came from by showing the document, page, evidence, and verification result.
- Human review ensures that AI assists the insurance advisor rather than replacing the advisor's final judgement.

The PowerPoint export is also separated from the AI-generation stage. It formats the already generated and reviewed workflow results instead of generating new policy information during export.

8. Limitations and Future Scope

The quality of the recommendation depends on the completeness and quality of the supplied policy documents. Some information, such as final premiums, quotations, exclusions, or client-specific commercial terms, may require additional insurer confirmation.

The current prototype can also be affected by external LLM API rate and token limits when a large amount of policy evidence is processed simultaneously.

Future improvements could include support for more insurers, improved document retrieval, larger document collections, richer policy comparison, persistent advisor sessions, authentication, and integration with insurer or broker systems.

9. Conclusion

INSUREAI demonstrates how generative AI can assist insurance advisors while maintaining evidence traceability and human oversight.

The prototype combines document retrieval, AI-based recommendation generation, claim-level verification, advisor approval, and automated presentation/report generation into a single workflow.

The result is a practical system that helps transform insurance policy documents into a structured client pitch while keeping important claims linked to supporting evidence.